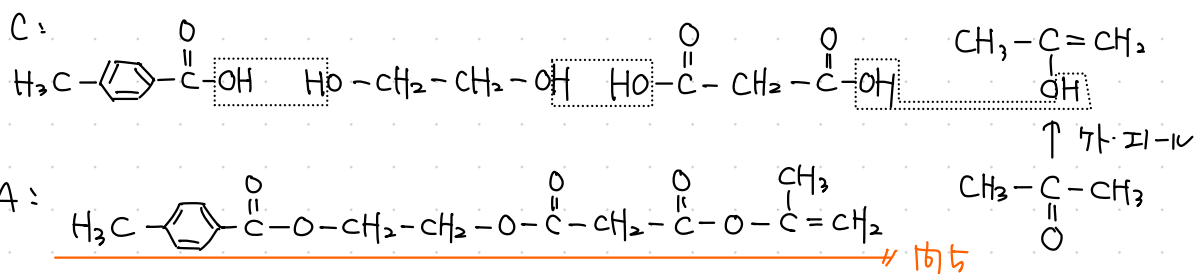
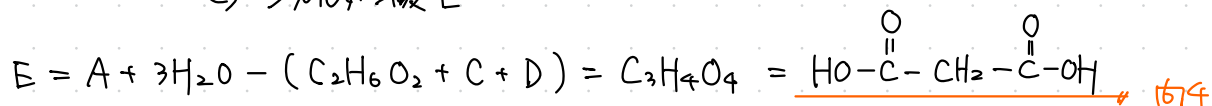
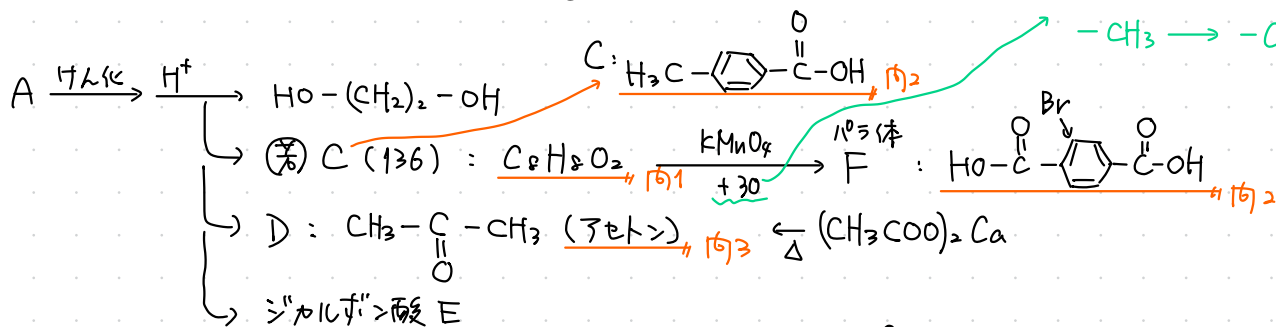


1 THU 2025

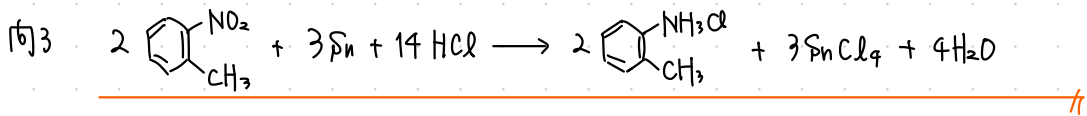
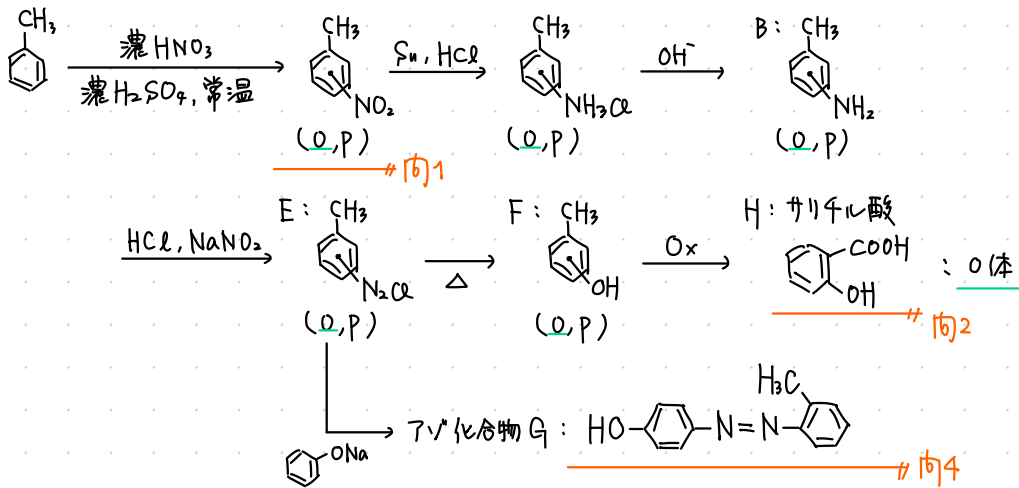
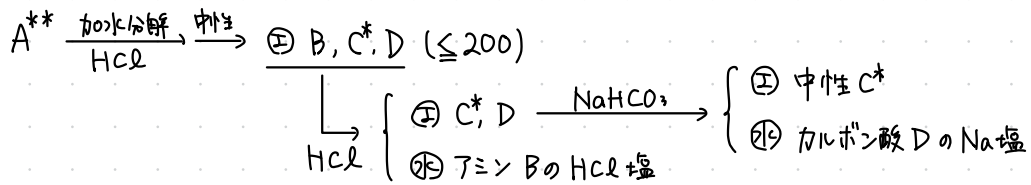
エステル A : $C_{16}H_{18}O_6$ $U=8 = -\text{C}(=\text{O})-\text{O}- \times 3 + \text{C}_6\text{H}_5 \times 1 + 1$

$\xrightarrow{\text{KMnO}_4}$ $2^\circ + 30$ は
 $-\text{CH}_3 \rightarrow -\text{COOH}$

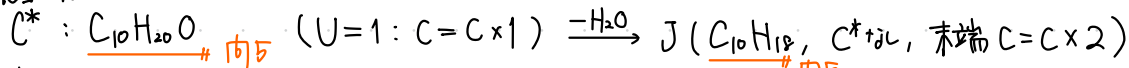


2 THU 2024

$A^{**} : C_x H_y O_z N_w$, C^* 2017 4級C原子2117個.

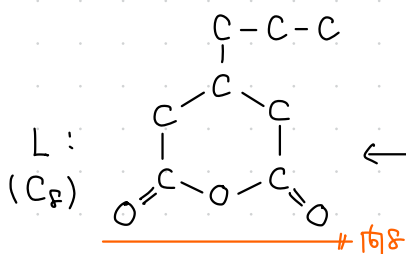
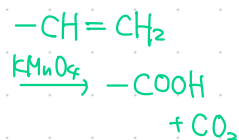
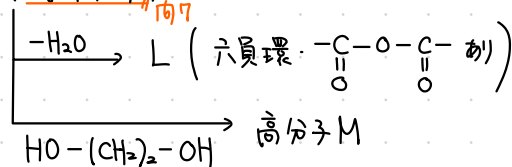
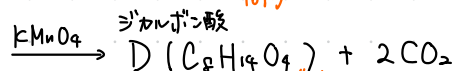


1級
アルコール

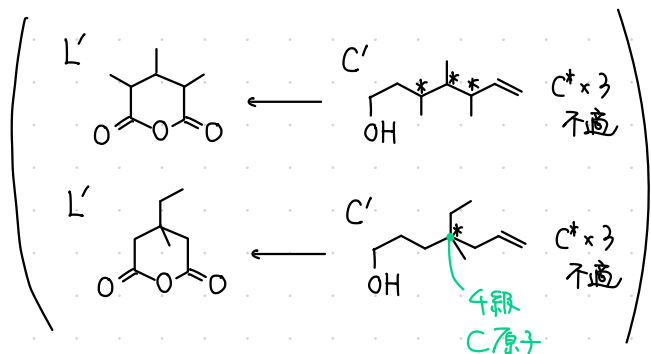
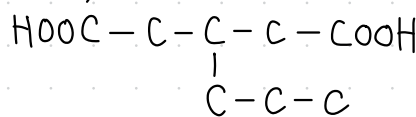


↓ H_2

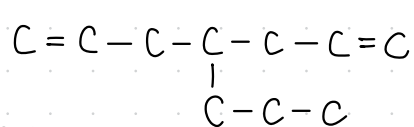
K(C^* なし = 対称面あり)



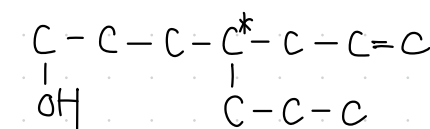
D(C_8)



← J(C_{10})

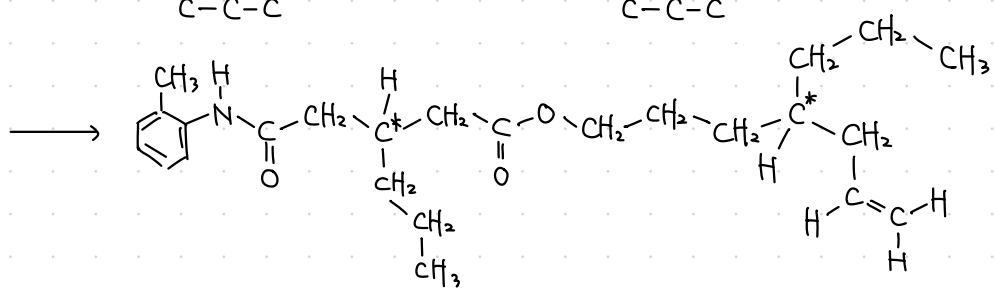
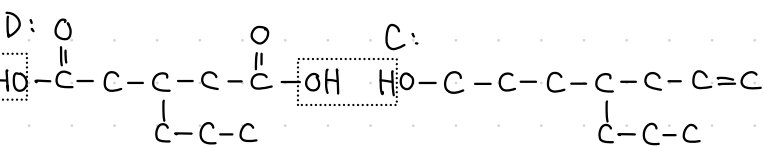
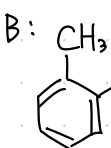
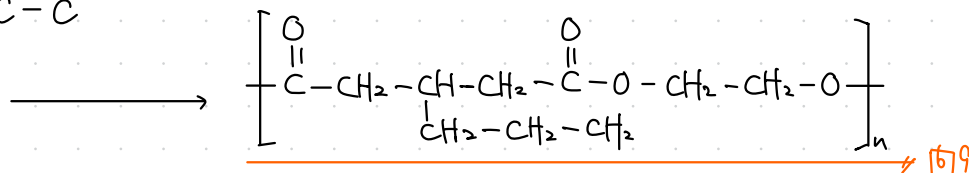
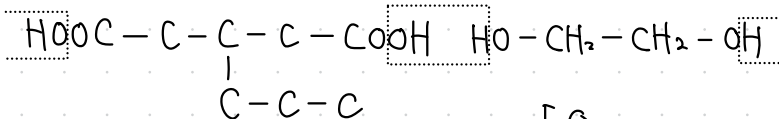


← C(C_{10})



108

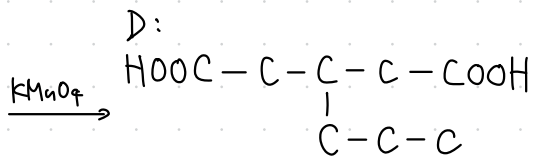
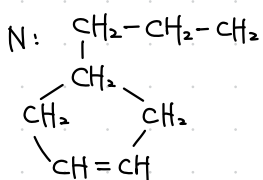
D:



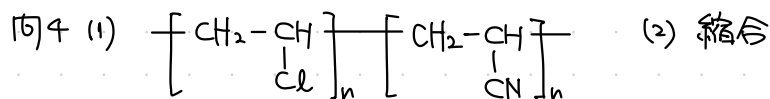
1012

10 2.4 × 10³ g

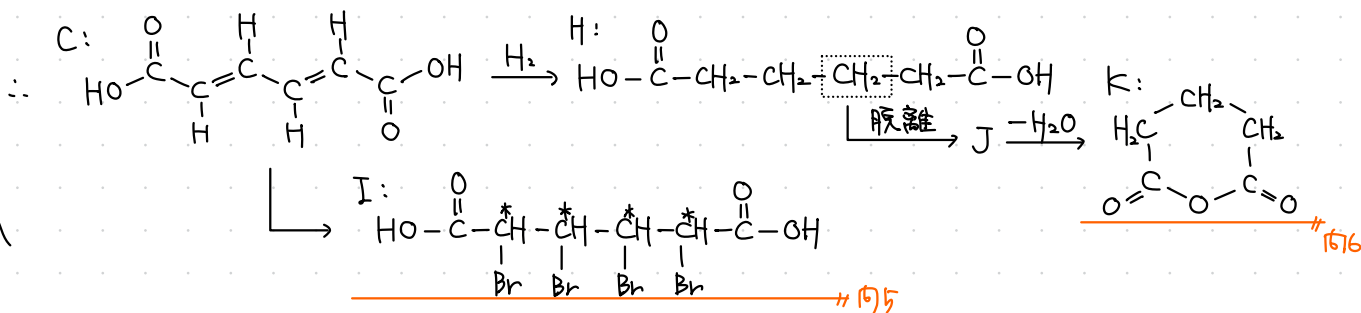
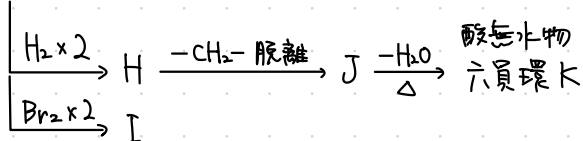
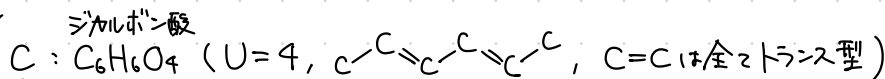
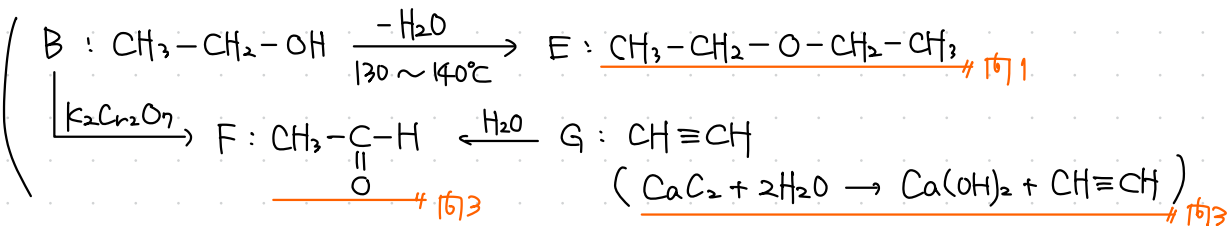
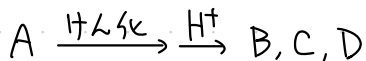
11 C_8H_{14} U=2



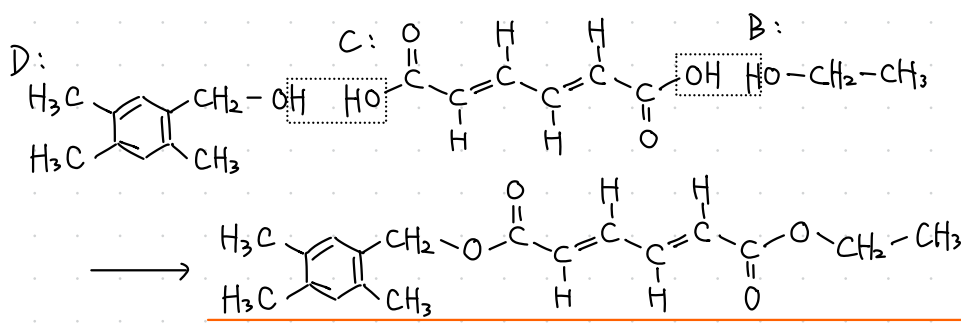
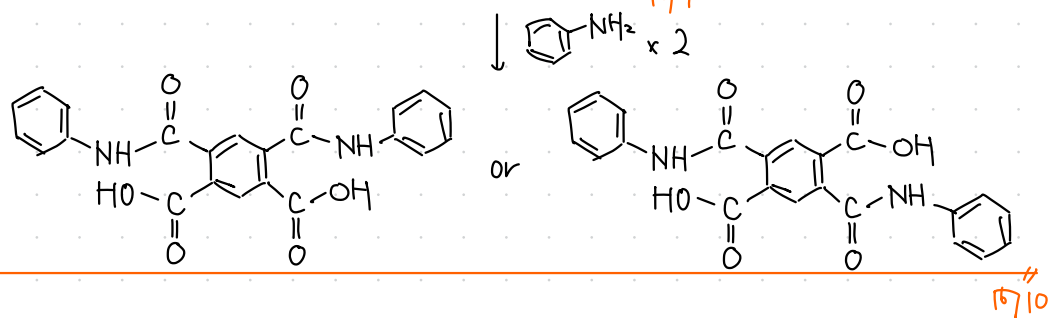
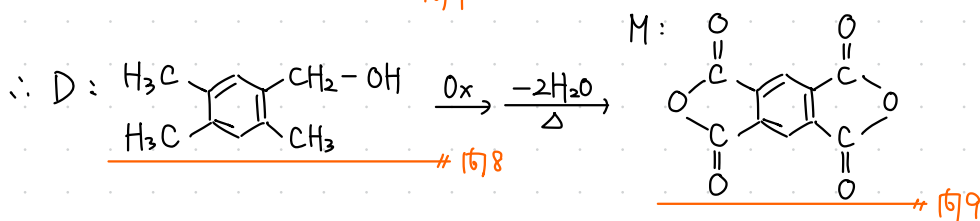
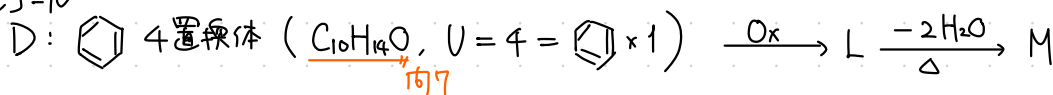
3 THU 2023



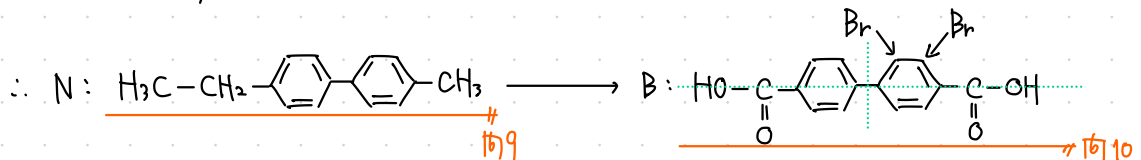
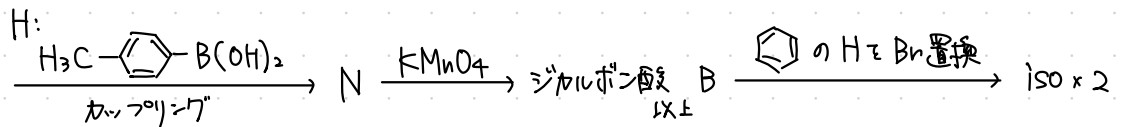
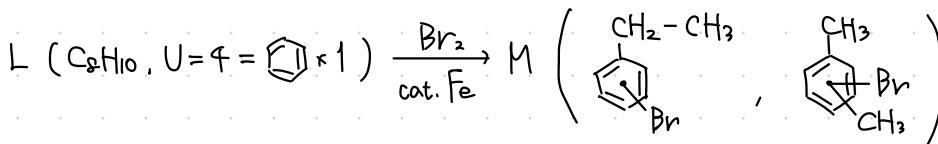
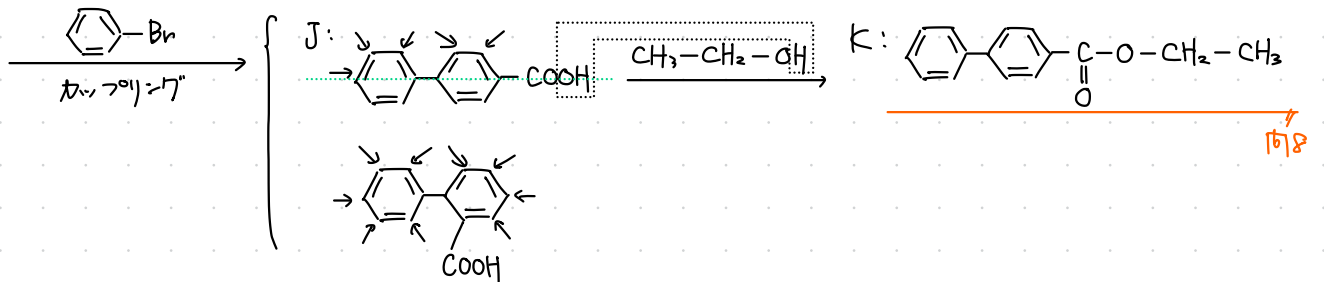
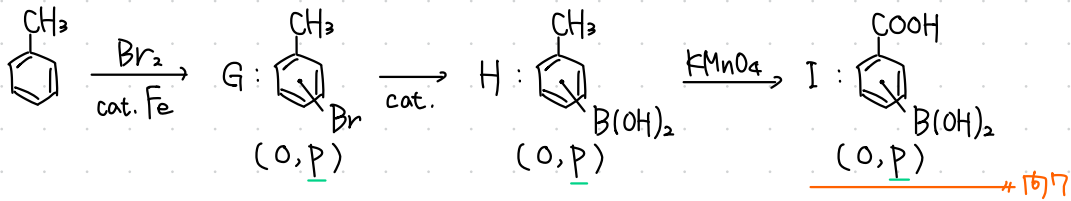
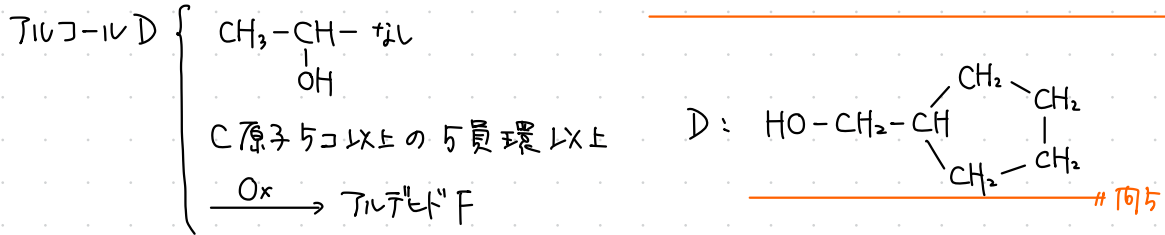
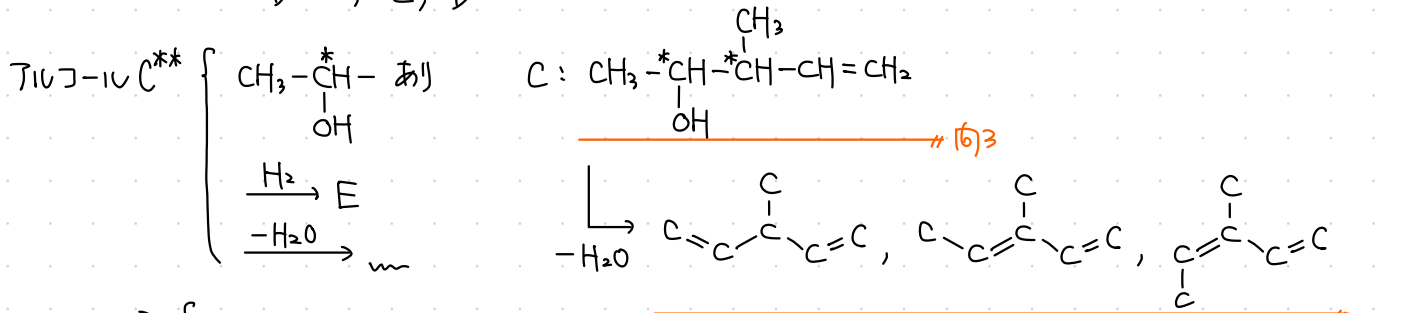
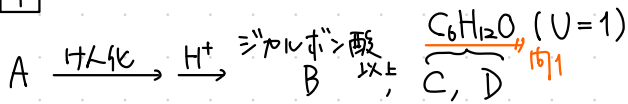
(3) $2.00 \times 10^2 : 1.00 \times 10^2$



第1級
アルコ-ル

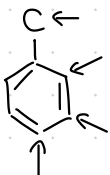


4



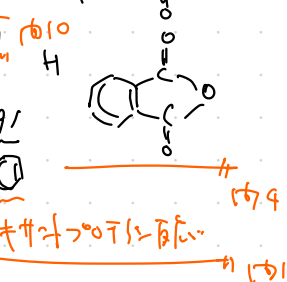
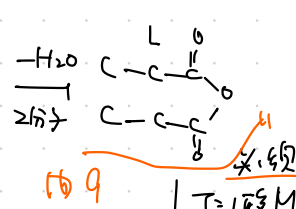
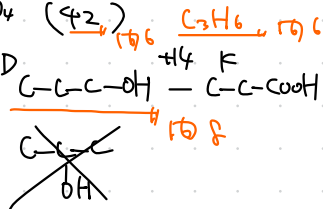
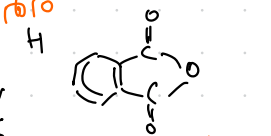
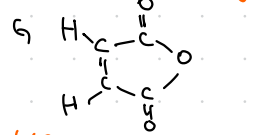
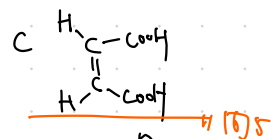
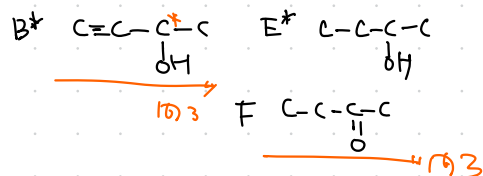
162 田舎

166

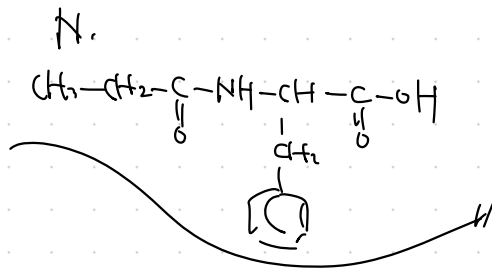


4

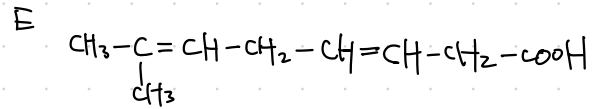
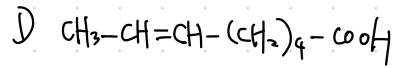
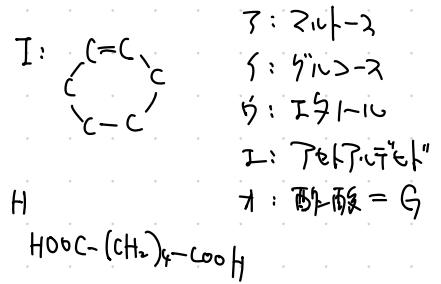
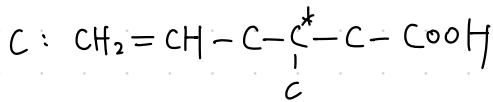
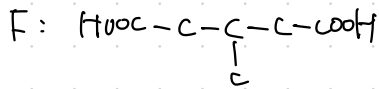
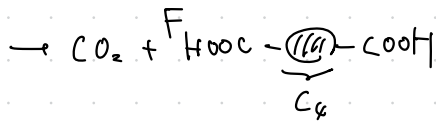
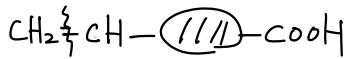
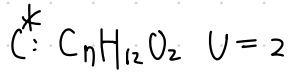
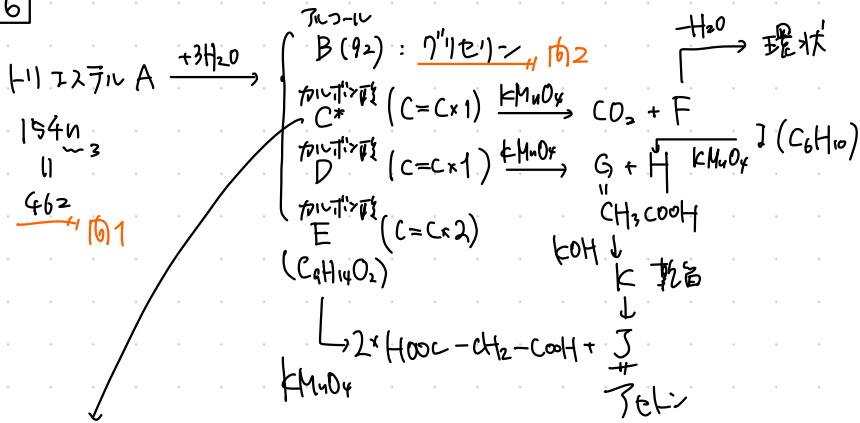
$\begin{array}{cccc} \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & \uparrow & & \uparrow & & \\ & & \textcircled{2} & & \textcircled{1} & & \end{array}$



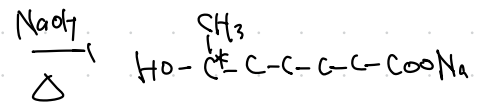
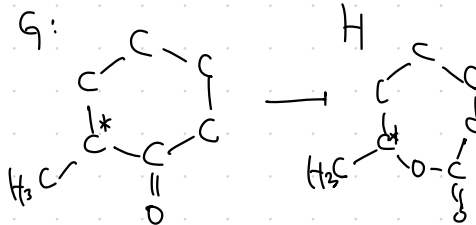
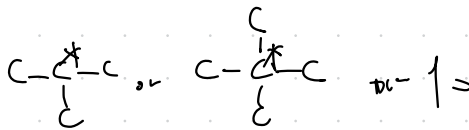
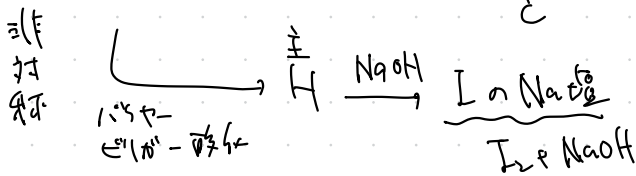
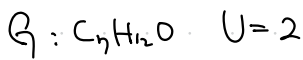
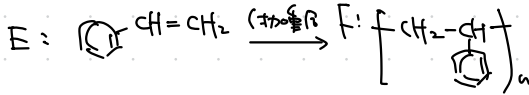
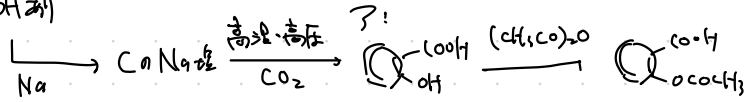
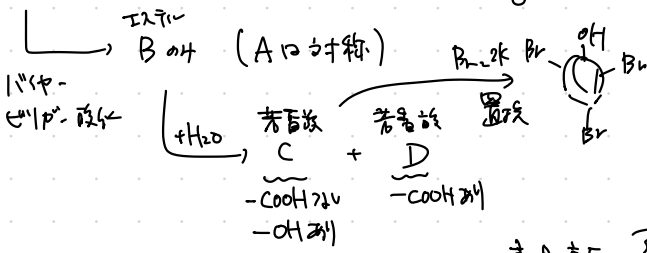
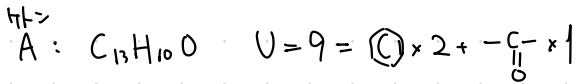
- N = 74 + 91
- CH₂ - 



6

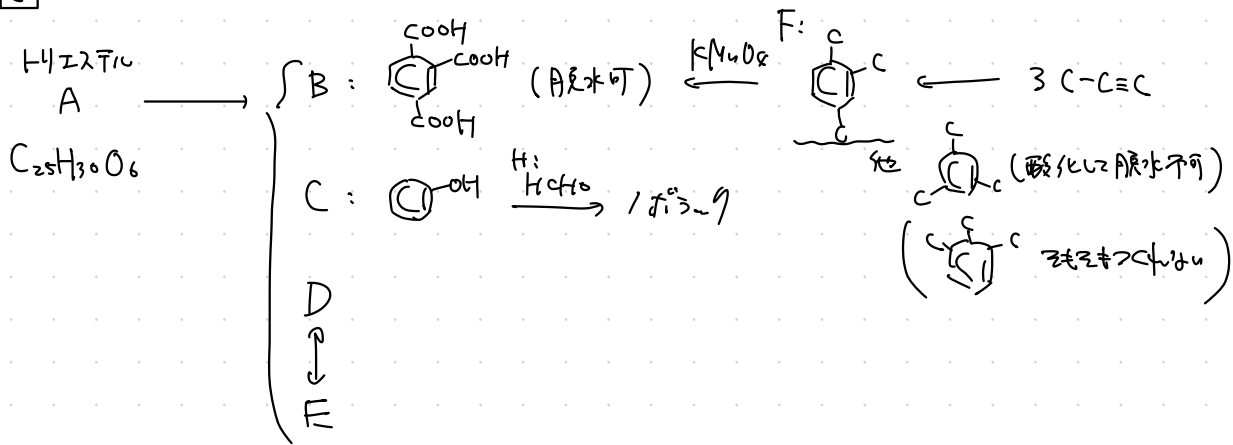


7

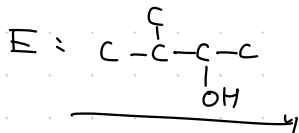
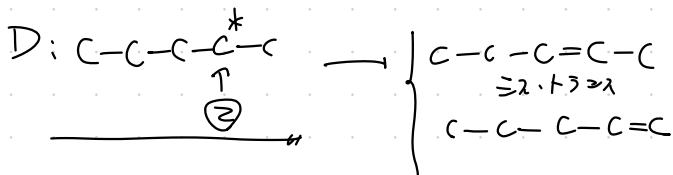


ア: e i: c り: h i: b

8



$$\text{D, E: } \frac{\text{C}_{25}\text{H}_{30}\text{O}_6 + 3\text{H}_2\text{O} - \text{C}_9\text{H}_6\text{O}_6 - \text{C}_6\text{H}_6\text{O}}{2} = \text{C}_5\text{H}_{12}\text{O}$$



9

$(C_2H_5O)_n$ について,

$$\frac{5n}{H数} \leq \frac{2 \cdot 2n + 2}{最大H数} \quad \therefore n \leq 2 \quad (H数は偶数), n=2.$$

$$\begin{aligned} &\rightarrow A \sim E \quad \left\{ \begin{array}{l} A, C, D, E : -OH \times 2 \\ B : -OH \times 1 + -O- \times 1 \end{array} \right. \\ &\quad (C_4H_{10}O_2) \\ &\quad U=0 \end{aligned}$$

